RAJALAKSHMI ENGINEERING COLLEGE

Mahalakshmi B 250701401-2025-CSE

M2

Started on	Thursday, 27 August 2026, 3:56 PM
State	Finished
Completed on	Thursday, 27 August 2026, 3:56 PM
Time taken	28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Divide and Conquer approach to find the index of the first 0
int findFirstZero(int arr[], int low, int high) {
    if (high >= low) {
        int mid = low + (high - low) / 2;

        // Check if mid element is the first 0
        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
            return mid;
        }

        // If mid element is 1, the first 0 must be in the right half
        if (arr[mid] == 1) {
            return findFirstZero(arr, mid + 1, high);
        }

        // Otherwise, the first 0 is in the left half
        return findFirstZero(arr, low, mid - 1);
    }
    return -1; // Return -1 if no 0 is present
}

int countZeroes(int arr[], int n) {
    int firstZero = findFirstZero(arr, 0, n - 1);

    if (firstZero == -1) {
        return 0;
    }

    return n - firstZero;
}

int main() {
    int m;
    if (scanf("%d", &m) != 1) return 0;

    int arr[m];
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    printf("%d\n", countZeroes(arr, m));

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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